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COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

40/25 +10 = 50

Midsemester Exam #2

Only one sheet of paper 5 x 7 is allowed

***Duration: 55 minutes* --- March 9, 2020**

Problem 1: (50 points)

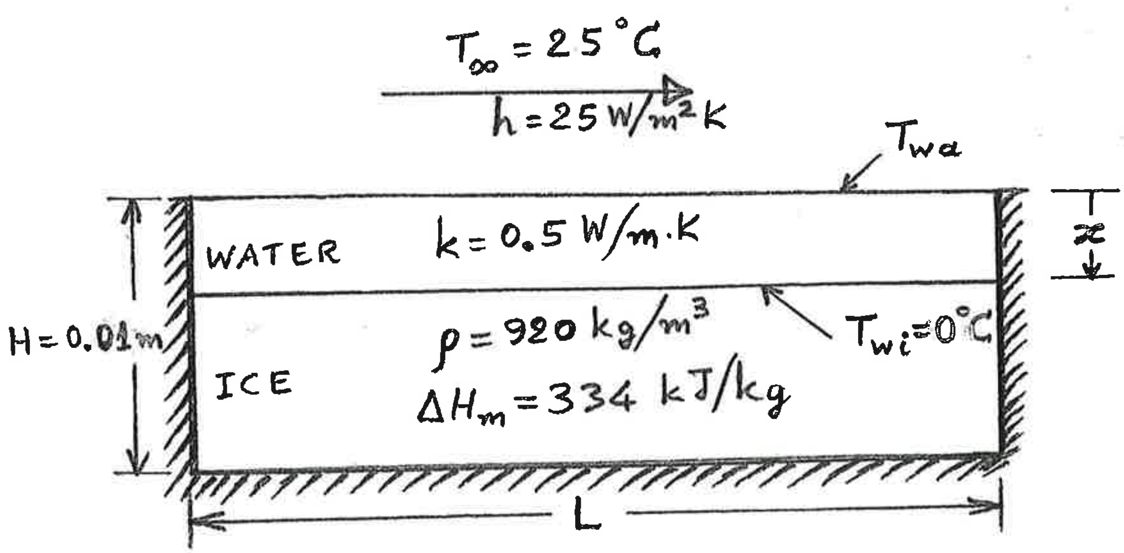
A slab of ice in a container of dimensions L, L, and H is placed on a well-insulated pad. At its top surface, initially the ice is exposed to ambient air for which T_∞ = 25 °C and the convection coefficient is 25 W/m^2·K. The ice is melting and the water formed acts as an additional resistance to the heat transfer. Assume that there is no storage of energy in the water being formed and that the water temperature profile is linear. The ice temperature remains at 0 °C and H = 10 mm. If the density and latent heat of fusion of ice are 920 kg/m^3 and 334 kJ/kg, respectively, and the thermal conductivity of water is 0.5 W/m·K, then:

10/50

a) By performing energy balance(s) over appropriate control element(s) determine the position of the water/ice interface as a function of time.

b) Determine the time required for the ice to melt. How does this relate to the time found in the case of no water-layer resistance?

Bonus: (25 points) Determine how the water-air interface temperature, T_{wa}, changes with time. What is the value of T_{wa} when the ice has melted?



Since this is a Slab in the X-Y direction, I will use Cartesian Coordinates and treat this problem with no internal generation and constant thermal conductivity

Balance: $\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$ $\quad \dot{q}_{conv} = \frac{d}{dt}(m \Delta H_m)$

$T_\infty = 25°C$
$K = .5 \frac{W}{m \cdot K}$
$H = 10 \text{ mm} = .01 \text{ m}$
$T = 0°C$
$h = 25 \frac{W}{m^2 K}$

Ice: $\rho = 920 \frac{Kg}{m^3}$
$\Delta H_m = 334 \frac{KJ}{Kg}$

Initial Condition: $T(x,0) = T_i$

X

Boundary Conditions: $\left.\frac{\partial T}{\partial x}\right|_{x=0} = 0$

$-K \left.\frac{\partial T}{\partial x}\right|_{x=L} = h(T(L,t) - T_\infty)$

$\alpha \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t} \rightarrow \boxed{\alpha \frac{\partial^2 T}{\partial T} = \frac{\partial x^2}{\partial t}}$

Balance: $-\dot{E}_{out} = E_{Acc}$

$-hA_s(T - T_\infty) = \rho V c \frac{dT}{dt}$

$Bi = \frac{hL}{K}$

* We can rewrite K to be α

I.C: at $t=0$, $T = T_i$

$t = \frac{\rho V c}{h A_s} \ln \frac{\theta_i}{\theta}$

B.C: at $X = L$, $-K \frac{\partial T}{\partial x} = h(T - T_\infty)$

at $X = -L$, $-K \frac{\partial T}{\partial x} = -h(T - T_\infty)$

at $X = 0$, $\frac{\partial T}{\partial x} = 0$

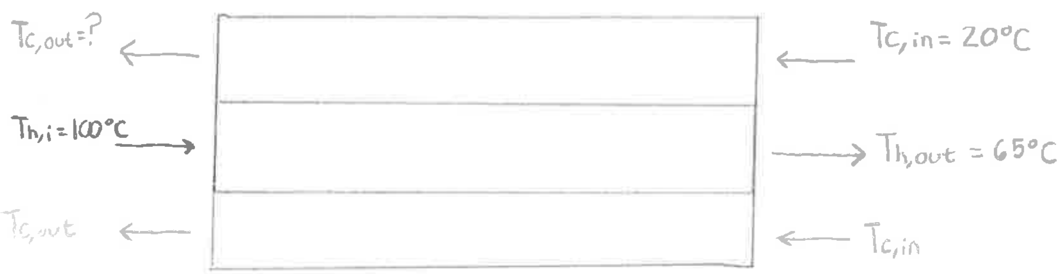
X

$\boxed{t = \frac{\rho V_c}{h A_s} \ln \frac{\theta_i}{\theta}}$ $\quad$ * $\frac{\theta_i}{\theta} = \frac{T - T_\infty}{T_i - T_\infty}$

Problem 2: (75 points)

A fluid at 100 °C is to be cooled by a different stream of the same fluid at 20 °C. A heat exchanger is available for parallel or counter flow. When the heat exchanger is used with parallel flow, the outlet temperature of the hot stream is 65 °C. If the mass flow rates in the hot and cold streams are equal to each other and the heat capacity of the fluid is temperature independent, determine:

(a) **(30 points)** The outlet temperature of the cold stream with parallel flow in the heat exchanger, and
(b) **(45 points)** The outlet temperatures of the cold and hot streams when a counter flow set up is used.



I will treat this problem as a parallel heat exchanger and write the energy balance that will help determine $T_{c,out}$: $q = UA\Delta T_{lm}$, where $\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)}$. We can use the NTU method since the inlet temperatures are know but outlets are not known.

$$NTU = \frac{UA}{C_{min}},$$ Parallel flow NTU: $$NTU = \frac{-\ln[1-\varepsilon(1+C_r)]}{1+C_r}$$

$$\varepsilon = \frac{1-\exp[-NTU(1+C_r)]}{1+C_r} = \frac{q}{q_{max}}$$

$$q_{max} = C_{min}(T_{h,i} - T_{c,i})$$

$C_{min} = C_c$ or $C_h \rightarrow C_c = \dot{m}_c C_{p,c}$ or $C_h = \dot{m}_h C_{p,h}$

Since $\dot{m}_c = \dot{m}_h$ and heat capacity is temperature independent: $C_r = \frac{C_{min}}{C_{max}} = \frac{\cancel{\dot{m}_h} C_{p,h}}{\cancel{\dot{m}_c} C_{p,c}} = 1$

$$\Delta T_{lm} = \frac{T_{h,out} - T_{c,in}}{\ln\left(\frac{T_{h,out}}{T_{c,in}}\right)} = \frac{(65-20)^\circ C}{\ln\left(\frac{65}{20}\right)^\circ C} = 38.18\ ^\circ C$$

$q_1 = 38.18\ UA \rightarrow$ Some value [=] W treated as a constant C_1

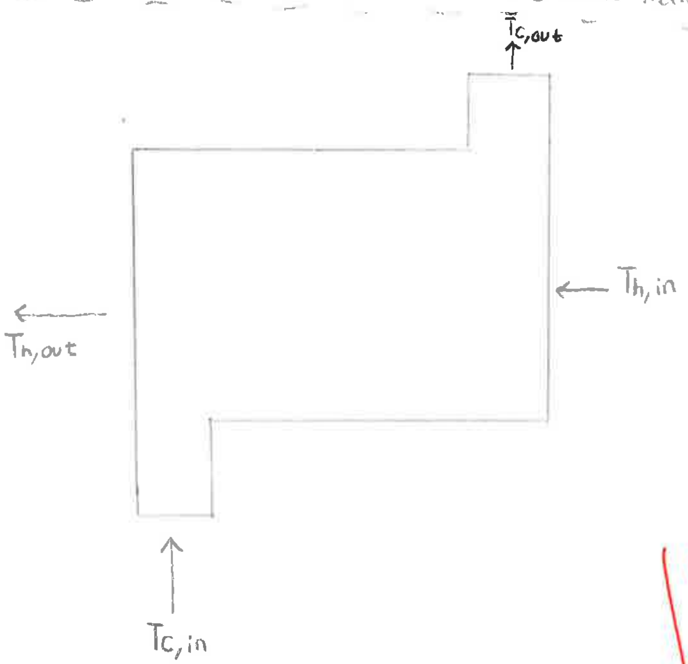
• q_1 will be represented with log mean temperature as a function of $T_{h,out}$ and $T_{c,in}$.

$$\Delta T_{lm} = \frac{T_{h,i} - T_{c,out}}{\ln\left(\frac{T_{h,i}}{T_{c,out}}\right)} = \frac{100 - T_{c,out}}{\ln\left(\frac{100}{T_{c,out}}\right)}$$

$$\boxed{C_1 = UA\left(\frac{100 - T_{c,out}}{\ln\left(\frac{100}{T_{c,out}}\right)}\right)}$$

From our C_1 equation, we can do some algebraic manipulations and solve for $T_{c,out}$

b.



For a counter flow heat exchanger:

$$NTU = \frac{1}{Cr-1}\ln\left(\frac{\varepsilon-1}{\varepsilon Cr-1}\right)$$

$$Cr = 1 \rightarrow C_{min} = C_{max}$$

$$\varepsilon = \frac{1-\exp[-NTU(1-Cr)]}{1-Cr\exp[-NTU(1-Cr)]}$$

We can follow a similar process by getting q and back calculating to find $T_{c,out}$ and $T_{h,out}$ using ΔT_{lm}.

NOT NECESSARILY NEEDED

TABLE 11.3 Heat exchanger effectiveness relations [5]

Flow Arrangement	Relation		
Parallel flow	$\varepsilon = \dfrac{1 - \exp[-\text{NTU}(1 + C_r)]}{1 + C_r}$		(11.28a)
Counterflow	$\varepsilon = \dfrac{1 - \exp[-\text{NTU}(1 - C_r)]}{1 - C_r \exp[-\text{NTU}(1 - C_r)]}$	$(C_r < 1)$	
	$\varepsilon = \dfrac{\text{NTU}}{1 + \text{NTU}}$	$(C_r = 1)$	(11.29a)
Shell-and-tube			
One shell pass (2, 4, . . . tube passes)	$\varepsilon_1 = 2\left\{1 + C_r + (1 + C_r^2)^{1/2} \times \dfrac{1 + \exp[-(\text{NTU})_1(1 + C_r^2)^{1/2}]}{1 - \exp[-(\text{NTU})_1(1 + C_r^2)^{1/2}]}\right\}^{-1}$		(11.30a)
n shell passes ($2n$, $4n$, . . . tube passes)	$\varepsilon = \left[\left(\dfrac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1}\right)^n - 1\right]\left[\left(\dfrac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1}\right)^n - C_r\right]^{-1}$		(11.31a)
Cross-flow (single pass)			
Both fluids unmixed	$\varepsilon = 1 - \exp\left[\left(\dfrac{1}{C_r}\right)(\text{NTU})^{0.22}\{\exp[-C_r(\text{NTU})^{0.78}] - 1\}\right]$		(11.32)
C_{max} (mixed), C_{min} (unmixed)	$\varepsilon = \left(\dfrac{1}{C_r}\right)(1 - \exp\{-C_r[1 - \exp(-\text{NTU})]\})$		(11.33a)
C_{min} (mixed), C_{max} (unmixed)	$\varepsilon = 1 - \exp(-C_r^{-1}\{1 - \exp[-C_r(\text{NTU})]\})$		(11.34a)
All exchangers ($C_r = 0$)	$\varepsilon = 1 - \exp(-\text{NTU})$		(11.35a)

TABLE 11.4 Heat exchanger NTU relations

Flow Arrangement	Relation		
Parallel flow	$\text{NTU} = -\dfrac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$		(11.28b)
Counterflow	$\text{NTU} = \dfrac{1}{C_r - 1}\ln\left(\dfrac{\varepsilon - 1}{\varepsilon C_r - 1}\right)$	$(C_r < 1)$	
	$\text{NTU} = \dfrac{\varepsilon}{1 - \varepsilon}$	$(C_r = 1)$	(11.29b)
Shell-and-tube			
One shell pass (2, 4, . . . tube passes)	$(\text{NTU})_1 = -(1 + C_r^2)^{-1/2}\ln\left(\dfrac{E - 1}{E + 1}\right)$		(11.30b)
	$E = \dfrac{2/\varepsilon_1 - (1 + C_r)}{(1 + C_r^2)^{1/2}}$		(11.30c)
n shell passes ($2n$, $4n$, . . . tube passes)	Use Equations 11.30b and 11.30c with $\varepsilon_1 = \dfrac{F - 1}{F - C_r}$ $\quad F = \left(\dfrac{\varepsilon C_r - 1}{\varepsilon - 1}\right)^{1/n}$ $\quad \text{NTU} = n(\text{NTU})_1$		(11.31b, c, d)
Cross-flow (single pass)			
C_{max} (mixed), C_{min} (unmixed)	$\text{NTU} = -\ln\left[1 + \left(\dfrac{1}{C_r}\right)\ln(1 - \varepsilon C_r)\right]$		(11.33b)
C_{min} (mixed), C_{max} (unmixed)	$\text{NTU} = -\left(\dfrac{1}{C_r}\right)\ln[C_r \ln(1 - \varepsilon) + 1]$		(11.34b)
All exchangers ($C_r = 0$)	$\text{NTU} = -\ln(1 - \varepsilon)$		(11.35b)